

Computability and Logic August 2023

Given is a vocabulary with function symbols $\{f, g\}$ and predicate symbols $\{A, B\}$. We use $\{x, y, z\}$ as variables.

Which of the following is a valid application of the $(\forall e)$ -rule in natural deduction?

Choose one answer

- ☐
$$\frac{\forall x(B(x) \rightarrow \exists y A(x, y))}{B(f(y)) \rightarrow \exists y A(f(y), y)} (\forall e)$$
- ☐
$$\frac{\forall x(B(x) \rightarrow \exists y A(x, y))}{B(B(z)) \rightarrow \exists y A(B(z), y)} (\forall e)$$
- ☐
$$\frac{\forall x(B(x) \rightarrow \exists y A(x, y))}{B(f(g(z))) \rightarrow \exists y A(f(g(z)), y)} (\forall e)$$
- ☐
$$\frac{\forall x(B(x) \rightarrow \exists y A(x, y))}{B(f(f(z))) \rightarrow \exists y A(g(g(z)), y)} (\forall e)$$

Which of the following problems on Context Free Grammars is undecidable?
(below, w is a word and G_1, G_2 are context-free grammars)

Choose one answer

- ☐ Given G_1, G_2, w , is $w \in \mathcal{L}(G_1) \cap \mathcal{L}(G_2)$?
- ☐ Given G_1, G_2 , is $\mathcal{L}(G_1) \cup \mathcal{L}(G_2) = \emptyset$?
- ☐ Given G_1, G_2 , is $\mathcal{L}(G_1) \cap \mathcal{L}(G_2) = \emptyset$?
- ☐ Given G_1, G_2, w , is $w \in \mathcal{L}(G_1) \cup \mathcal{L}(G_2)$?

Which of the following formulas in propositional logic are satisfiable?

Choose one or more answers

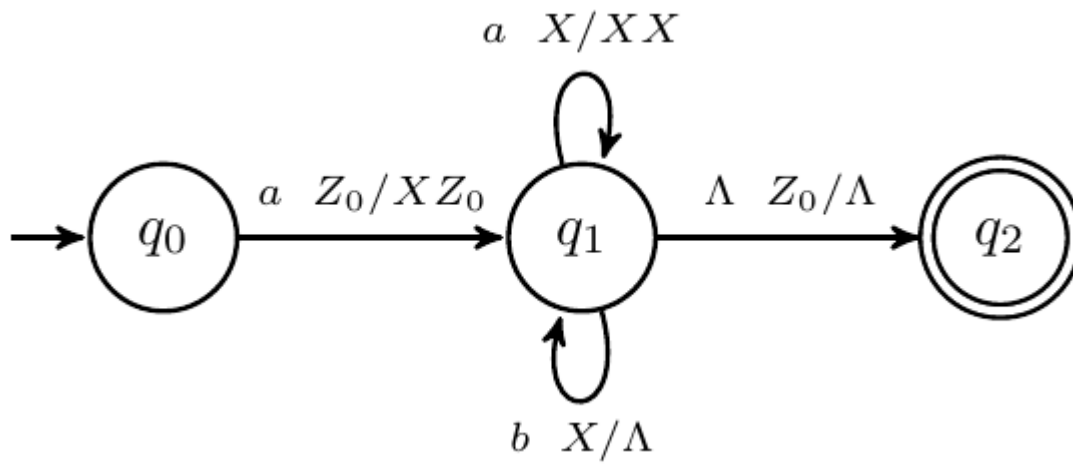
☐ $(p \vee q) \wedge (\neg p \wedge \neg q)$

☐ $(p \vee q) \wedge (\neg p \vee \neg q)$

☐ $(p \vee q) \wedge \neg(p \vee q)$

☐ $(p \vee q) \wedge \neg(p \wedge q)$

Consider the following PDA:



Which of the following words is accepted (by final state) by this PDA?

Choose one answer

- ☐ $ababab$
- ☐ $aaabb$
- ☐ $aabbba$
- ☐ $aaabbb$

Which of the following formulas in predicate logic are a tautology?

Choose one or more answers

☐ $\exists x \forall y (P(x) \rightarrow P(y))$

☐ $(\exists x P(x) \wedge \exists x Q(x)) \rightarrow \exists x (P(x) \wedge Q(x))$

☐ $\forall x \exists y (P(x) \rightarrow P(y))$

☐ $\exists x (P(x) \wedge Q(x)) \rightarrow (\exists x P(x) \wedge \exists x Q(x))$

Consider the following proof in natural deduction for propositional logic:

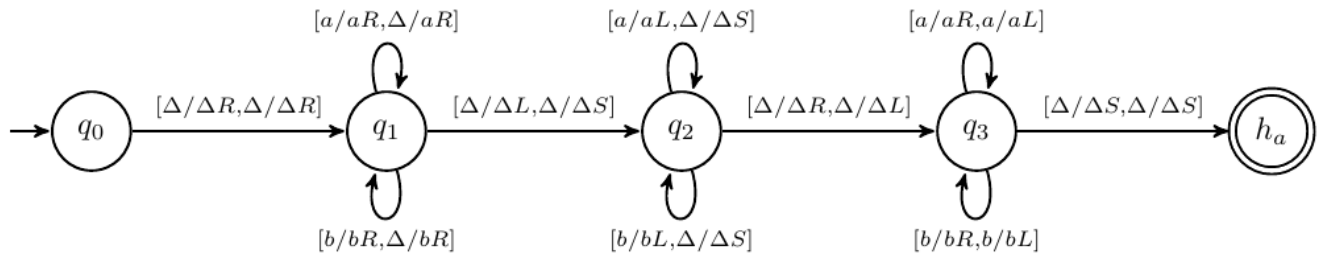
1. $\neg(p \wedge q)$
2. p
3. q
4. $p \wedge q$
5. $\perp$
6. r
7. $q \rightarrow r$
8. $p \rightarrow q \rightarrow r$
9. $\neg(p \wedge q) \rightarrow (p \rightarrow q \rightarrow r)$

Which of the following proof rules are used in this proof? (select all correct answers)

Choose one or more answers

- ☐ $\wedge$ -elimination
- ☐ $\rightarrow$ -elimination
- ☐ $\rightarrow$ -introduction
- ☐ $\neg$ -elimination
- ☐ $\neg$ -introduction
- ☐ $\wedge$ -introduction

What does the following multi-tape Turing Machine achieve?



(below, w^R denotes the reverse of word w)

Choose one answer

- ☐ It *decides* the language $\{w \in \{a, b\}^* \mid w = w^R\}$.
- ☐ It *decides* the language $\{ww \mid w \in \{a, b\}^*\}$.
- ☐ It *accepts* the language $\{w \in \{a, b\}^* \mid w = w^R\}$, but it doesn't *decide* it.
- ☐ It *accepts* the language $\{ww \mid w \in \{a, b\}^*\}$, but it doesn't *decide* it.

Which of the following rules can appear in a context-free grammar in Chomsky Normal Form?

(alphabet $\Sigma = \{0, 1\}$, non-terminals $\Gamma = \{S, A, B\}$)

Choose one or more answers

☐ $S \rightarrow A$

☐ $S \rightarrow AB$

☐ $S \rightarrow 0S1$

☐ $S \rightarrow 0$

Three of the following regular expressions are equivalent, but one of them generates a different language. Which one is different?

Choose one answer

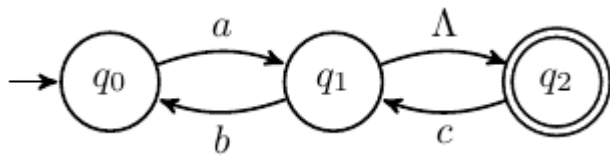
- ☐ $b(ba)^*$
- ☐ $bb(ab)^*a$
- ☐ $b + b(ba)^*$
- ☐ $b + bb(ab)^*a$

Which statement(s) on the decidability of logical systems are correct?

Choose one or more answers

- ☐ *Provability* is decidable for predicate logic.
- ☐ *Provability* is decidable propositional logic.
- ☐ *Validity* is decidable for predicate logic.
- ☐ *Validity* is decidable for the logic of arithmetic.

Consider the following Nondeterministic Finite Automaton M:



Which of the following words are accepted by M?

Choose one or more answers

☐ *acccba*

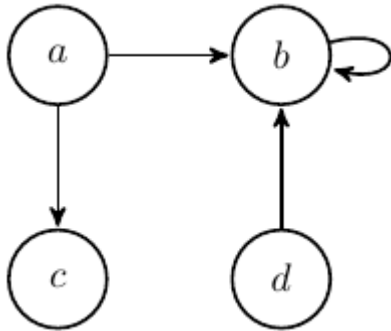
☐ *abccca*

☐ *accbab*

☐ *acbacc*

We consider predicate logic with a binary relation symbol T . We look at a particular model with universe $A = \{a, b, c, d\}$.

The relation is interpreted as in the following diagram, where $T(x, y)$ holds if x has an arrow to y .



Which of the following predicate-logic formulas holds in this model?

Choose one or more answers

- ☐ $\exists x \forall y T(y, x)$
- ☐ $\forall x \exists y (T(x, y) \vee T(y, x))$
- ☐ $\exists x \forall y \neg T(y, x)$
- ☐ $\forall x \forall y \forall z (T(x, y) \wedge T(x, z) \rightarrow y = z)$

Which of the following statements are correct?
(equivalent = accepts the same language)

Choose one or more answers

- ☐ Every non-deterministic *Push-Down Automaton* can be transformed to an equivalent deterministic *Turing Machine*.
- ☐ Every non-deterministic *Finite Automaton* can be transformed to an equivalent deterministic one.
- ☐ Every non-deterministic *Push-Down Automaton* can be transformed to an equivalent deterministic one.
- ☐ Every deterministic *Turing Machine* can be transformed to an equivalent *Push Down Automaton*

Which one of the following languages are Context-Free?
(the alphabet is $\{a,b,c\}$).

Choose one or more answers

- ☐ The set of words that have more a's than b's *or* more b's than c's.
- ☐ The set of words that have more a's than b's *and* more b's than c's.
- ☐ The set of words that have an equal number of a's, b's and c's.
- ☐ The set of all words of odd length with a "b" in the middle.

The following problems all take a Turing Machine M as input. To which problem can we apply Rice's theorem to conclude undecidability?

Choose one answer

- ☐ M has more than 10 states.
- ☐ M takes 10 steps on input Λ
- ☐ M accepts a word of length 10
- ☐ $\mathcal{L}(M)$ is a countable set

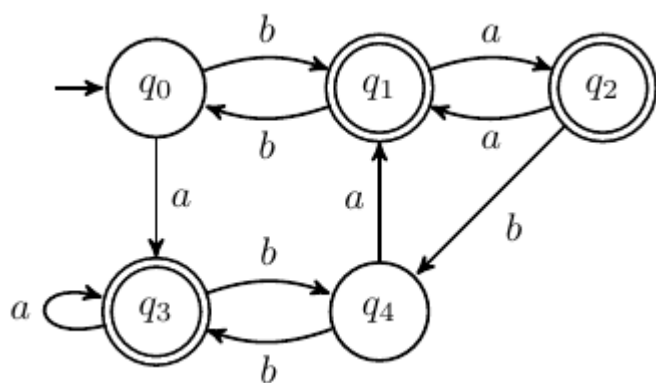
Consider the following dominoes for Post's Correspondence Problem:

$$x_1 = \begin{pmatrix} 00 \\ 0 \end{pmatrix}, x_2 = \begin{pmatrix} 00 \\ 0010 \end{pmatrix}, x_3 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Choose one answer

- ☐ This instance of PCP has solution x_2, x_3, x_1, x_1
- ☐ This instance of PCP has no solution at all
- ☐ This instance of PCP has solution x_2, x_3, x_1
- ☐ This instance of PCP has solution x_2, x_3, x_1, x_1, x_1

Consider the following DFA:



Which pair of states of this DFA are distinguishable?

Choose one answer

- ☐ q_1 is distinguishable from q_2
- ☐ q_0 is distinguishable from q_4
- ☐ q_0 is distinguishable from q_1
- ☐ q_2 is distinguishable from q_3

Which of the following Hoare triples is valid?

Choose one answer

- ☐ $\{x + y \leq 4\} \ z := x + y; x := 3 * z \ \{x \leq 15\}$
- ☐ $\{x + y \leq 6\} \ x := 3 * z; z := x + y \ \{x \leq 15\}$
- ☐ $\{x + y \leq 6\} \ z := x + y; x := 3 * z \ \{x \leq 15\}$
- ☐ $\{x + y \leq 4\} \ x := 3 * z; z := x + y \ \{x \leq 15\}$

Below, SAT is the satisfiability problem for propositional logic, and X is an arbitrary problem in P.

Which one of the following statements on polynomial-time reduction is NOT known to be true?

Choose one answer

- ☐ If $X \leq_P SAT$, then X is NP-complete.
- ☐ If $X \leq_P SAT$, then X is in NP.
- ☐ If $SAT \leq_P X$, then X is NP-complete.
- ☐ If $SAT \leq_P X$, then $P = NP$.

Given is a context-free grammar G that is *ambiguous*.

Which one of the following sentences can be concluded?

Choose one answer

- ☐ The language generated by G contains more than one word.
- ☐ Every word in the language of G has more than one derivation.
- ☐ A non-ambiguous grammar that generates the same language as G does not exist
- ☐ There exists a word that has more than one derivation tree in G .

Which of the following statements on the relation between a language and its complement is WRONG?

Choose one answer

- ☐ If a language L is regular, then its complement L' is context-free
- ☐ If a language L is recursive, then its complement L' is recursively enumerable
- ☐ If a language L is context-free, then its complement L' is context-free
- ☐ If a language L is context-free, then its complement L' is recursive

Consider the following truth tables, for propositional formulas A, B, C and D:

p	q	r	A	B	C	D
T	T	T	F	F	F	F
T	T	F	F	T	F	F
T	F	T	F	F	F	F
T	F	F	F	F	T	F
F	T	T	F	T	F	F
F	T	F	T	T	F	F
F	F	T	F	F	T	F
F	F	F	F	F	T	T

What is the correct truth table for the formula $\neg(q \rightarrow (p \wedge r))$?

Choose one answer

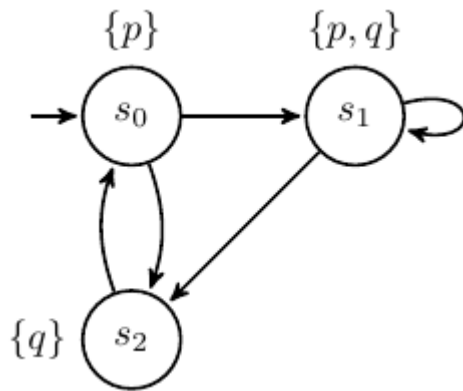
☐ B

☐ D

☐ A

☐ C

Consider the following Kripke structure:



Which of the following LTL properties hold on this model?

Choose one or more answers

- ☐ **FG** p
- ☐ **F** $\neg p$
- ☐ p **U** q
- ☐ **GF** q

Consider the following Context-Free Grammar in Chomsky Normal Form:

$$S \rightarrow AA \mid c$$

$$A \rightarrow SS \mid b$$

We use the CYK-algorithm to check if the word $bbcc$ is generated by this grammar.

A partially filled table is already shown.

How much entries in total will contain A or S , respectively?

A	?	?	?
	A	?	?
		S	?
			S

Choose one answer

- ☐ 4 times A and 4 times S
- ☐ 5 times A and 3 times S
- ☐ 4 times A and 3 times S
- ☐ 5 times A and 5 times S

Let e be a reasonable encoding for Turing Machines into words over the alphabet $\Sigma = \{0, 1\}$.

Which statement is true?

Choose one answer

- ☐ The language $\{e(M) \mid M \text{ accepts } 0110\}$ is not recursively enumerable
- ☐ The language $\{e(M) \mid M \text{ accepts } 0110\}$ is context-free (but not regular)
- ☐ The language $\{e(M) \mid M \text{ accepts } 0110\}$ is recursive (but not context-free)
- ☐ The language $\{e(M) \mid M \text{ accepts } 0110\}$ is recursively enumerable (but not recursive)

The following properties each define some language over the alphabet $\{a,b,c\}$. Which of these languages are regular?

Choose one or more answers

- ☐ The number of c's exceeds the sum of the number of a's and b's
- ☐ The symbols b and c do not occur next to each other.
- ☐ The total number of a's and b's is even, and the total number of b's and c's is odd.
- ☐ The length of the word is a power of 2.

Given are four grammars, with start symbol S. One of them generates a different language. Which one?

Choose one answer

- ☐
$$\begin{cases} S \rightarrow XY \\ X \rightarrow aX \mid \Lambda \\ Y \rightarrow bY \mid \Lambda \end{cases}$$
- ☐
$$\begin{cases} S \rightarrow XY \\ X \rightarrow aX \mid \Lambda \\ Y \rightarrow aYb \mid \Lambda \end{cases}$$
- ☐
$$\{ S \rightarrow aS \mid aSb \mid \Lambda$$
- ☐
$$\begin{cases} S \rightarrow aS \mid aT \mid \Lambda \\ T \rightarrow SB \\ B \rightarrow b \end{cases}$$

Consider an arbitrary regular language R and an arbitrary context-free language C . Which of the following statement can be concluded?
(X' denotes the complement of X).

Choose one answer

- ☐ $C' \cap R$ is regular
- ☐ $C' \cup R$ is context-free
- ☐ $C \cap R'$ is context-free
- ☐ $C \cup R'$ is regular

Consider the Hoare triple $\{P\} S \{Q\}$ under partial and total correctness. Which statement on this triple is correct?

Choose one answer

- ☐ Under total correctness, program S will terminate for every initial state
- ☐ Under total correctness, program S will terminate for every initial state that satisfy precondition P
- ☐ Under partial correctness, program S will terminate for every initial state that satisfy precondition P
- ☐ Under partial correctness, program S will terminate for every initial state

Consider decision problems P , Q and R and assume that $P \leq Q$ and $Q \leq R$. Which one of the following statements can be concluded?

Choose one answer

- ☐ If P is NP-hard then R is NP-hard
- ☐ If P is undecidable then R is undecidable
- ☐ If R is undecidable then P is undecidable
- ☐ If P is decidable then R is decidable

